Film Dosimetry

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1 Objective

- To establish a reliable dosimetric calibration for GAFchromic™ EBT3 film by relating film response to known absorbed doses under reference irradiation conditions.
- To determine an **unknown absorbed dose** from irradiated film by converting scanner pixel values (or net optical density) into dose using the established calibration curve.
- To perform a **spoke (starshot) test** using film for evaluation of radiation isocenter behavior and beam-axis convergence as part of machine quality assurance.
- To perform a **light and radiation field congruence test** (field coincidence test) to verify geometric agreement between optical field indicators and delivered radiation field.
- To carry out **patient-specific QA** by comparing measured planar film dose with treatment planning system (TPS) dose using profile comparison and γ-index analysis.
- To understand major sources of uncertainty in film dosimetry (scanner artifacts, film handling, post-irradiation growth, lot dependence, and setup reproducibility) and apply practical methods to reduce them.

2 Apparatus

- GAFchromic™ EBT3 Film
- Teletherapy Unit (Linear Accelerator or Co-60)
- Flatbed Document Scanner (e.g., EPSON Expression 10000–13000XL / equivalent) with 48-bit RGB acquisition
- Scanner software with all automatic corrections disabled (color restoration, sharpening, dust removal, auto-exposure)
- Computer workstation with Radiochromic.com workflow and/or film analysis software (for calibration, dose map generation, profiles, and γ-analysis)
- Uncompressed TIFF image format support for archival and analysis
- Water-equivalent slab phantom (solid water) for buildup and backscatter conditions
- Phantom positioning accessories: ruler, laser alignment, spirit level, and setup marks
- Film positioning frame to keep film centered and reproducibly oriented on scanner bed
- QA phantoms/fixtures for spoke (starshot) and field congruence tests

3 Theory

Radiation dosimetry is a critical aspect of radiotherapy and medical physics, ensuring that intended radiation doses are accurately and precisely delivered to specific target volumes while sparing healthy tissues. Film dosimetry stands out as one of the most powerful tools in this field due to its unparalleled high spatial resolution, which makes it ideal for measuring dose distributions with steep dose gradients, such as those found in Intensity Modulated Radiation Therapy (IMRT), Volumetric Modulated Arc Therapy (VMAT), and Stereotactic Radiosurgery (SRS) [1].

Historically and currently, film dosimetry predominantly involves two major categories of films: radiographic films and radiochromic films (RCF). Both films serve as dosimetric tools but operate on entirely fundamentally different chemical and physical principles.

3.1 Radiographic Film

Radiographic films have been utilized in medical imaging and dosimetry for over a century. The active sensing layer in a radiographic film consists of microscopic silver halide crystals (typically 95% silver bromide and 5% silver iodide) embedded in a gelatin matrix coated on a flexible polyester base. When ionizing radiation or light strikes the film, electrons are released, reducing silver ions (Ag^+) into metallic silver (Ag^0). This creates a "latent image" that is invisible to the naked eye. To render this image visible and suitable for dosimetric measurements, the film must undergo a rigorous wet chemical processing procedure involving a developer (to amplify the latent image), a fixer (to remove unexposed silver halides), washing, and drying.

3.2 Radiochromic Film (RCF)

Radiochromic film dosimetry is a more recent chemical dosimetry technique that eliminates the complexities of wet chemical processing. Instead of utilizing silver halides, the active radiation-sensitive medium in radiochromic films is composed of microcrystalline di-acetylene monomers (C_4H_2). When exposed to ionizing radiation, these highly unsaturated hydrocarbon monomers undergo a solid-state, chain-growth photo-polymerization process. This chemical reaction directly produces colored polymer chains, progressively turning the film darker (often a shade of blue or green depending on the specific model) as the absorbed dose increases. Because this color change occurs directly as a result of the irradiation, radiochromic films are self-developing [1].

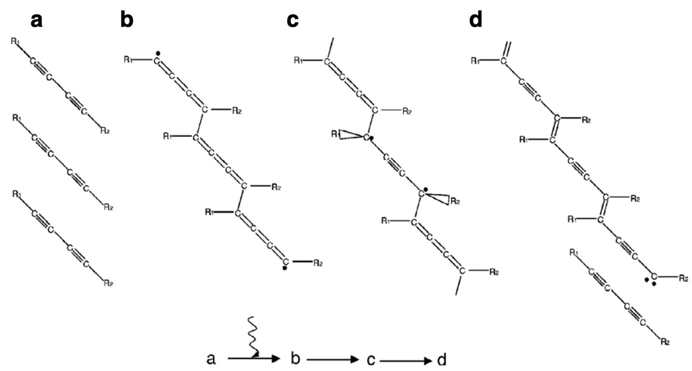


Figure 1: Diacetylene monomers, upon exposure to ionizing radiation, polymerize into (b) butatriene structure polymer; as the polymer chain grows, it rearranges via (c) an intermediate between butatriene structure and acetylene structure, into (d) acetylene structure polymer

4 Advantages of Radiochromic Film Over Radiographic Film

The transition from radiographic to radiochromic film in modern clinical dosimetry is driven by several significant physical and practical advantages of RCF over its predecessor:

- **Self-Developing Nature:** Because the radiochromic process relies on direct solid-state polymerization, it requires absolutely no chemical development, physical processing, or thermal treatment. This completely eliminates the need for darkrooms, complex processing machinery, and hazardous chemical waste disposal associated with radiographic films. It also removes the large dosimetric uncertainties intrinsic to temperature, chemical concentration, and agitation variations in wet chemical processing.
- **Tissue Equivalency:** Radiographic films possess a high effective atomic number ($Z_{eff} \approx 40 - 45$) due to the heavy silver halides, leading to substantial over-responses at low photon energies via the photoelectric effect. Conversely, radiochromic films are composed primarily of organic molecules containing hydrogen, carbon, and oxygen. Consequently, their effective atomic number ($Z_{eff} \approx 6.8$ to 7.5) is remarkably close to that of water and human tissue.
- **Insensitivity to Visible Light:** While radiographic films are extremely sensitive to visible light and must be handled under strict darkroom conditions, most modern radiochromic films are largely

insensitive to room light (incandescent light). They can be comfortably handled and prepared in normal ambient lighting, though prolonged exposure to direct sunlight or strong ultraviolet (UV) fluorescent sources should be avoided to prevent background photo-polymerization.

- **Water Resistance:** Many radiochromic film models (such as the EBT3) feature protective polyester laminations that restrict water ingress into the active layer. This allows the films to be fully immersed in water phantoms for direct absolute or relative depth-dose measurements, provided the cut edges are adequately sealed.
- **High Spatial Resolution:** RCF provides excellent spatial resolution limited only by the size of the microcrystalline grains (in the micrometer range) and the resolution capability of the readout scanner, making it exceptionally reliable for high-gradient stereotactic fields and micro-dosimetry.

5 Available RCF Models and Dosimetric Characteristics

Since the early 1990s, numerous models of radiochromic films have been developed and commercialized, predominantly under the GAFchromic^{TM} brand (Ashland Inc.). These models are tailored for different dose ranges and applications.

- **High Dose Films:** Early models like MD-55 and HD-810 were designed for high-dose ranges (10 Gy to 1000 Gy) used in industrial applications, blood irradiation, and brachytherapy.
- **External Beam Therapy (EBT) Films:** To match the required sensitivity for standard clinical external beam radiotherapy (cGy to Gy range), the EBT series was introduced.
- **GAFchromic EBT2 and EBT3:** EBT2 included a yellow dye in the active layer to reduce UV/visible light sensitivity and incorporated a marker dye to allow for active layer thickness corrections. EBT3 maintained the exact same active layer composition but introduced a symmetrical construction (matte polyester laminates on both sides to eliminate orientation dependence and Newton's rings artifacts during scanning).
- **EBT-XD:** Specifically engineered for the dose distributions of SRS and SBRT, the EBT-XD film offers optimal dose assessment for high dose deliveries (up to 40 Gy and beyond).

The dosimetric characteristics of radiochromic films determine their accuracy, reliability, and precision in clinical environments.

5.1 Energy and Linear-Energy Transfer (LET) Dependence

Because the chemical composition of the RCF active layer closely mimics human tissue and water ($Z_{eff} \approx 6.8$), it exhibits minimal energy dependence for megavoltage (MV) photon and electron beams. In ranges between 100 keV and 18 MeV, the dose response generally varies by less than a few percent. Even at lower kilovoltage energies (brachytherapy or orthovoltage), the energy dependence is considerably milder compared to radiographic film. However, radiochromic films do exhibit a Linear-Energy Transfer (LET) dependence. When irradiated by densely ionizing particles such as low-energy protons, alpha particles, or heavy carbon ions, the high local energy deposition along the track core rapidly saturates the local polymerization process. This phenomenon, often referred to as a "quenching effect," results in a decreased dosimetric efficiency and under-response of the film in the Bragg peak region of high-LET ion beams. Hence, rigorous LET-based corrections are necessary when utilizing RCF in hadron therapy dosimetry.

5.2 Dose-Rate Independence

A major prerequisite for modern relative dosimeters is the ability to handle the rapidly varying dose rates delivered by VMAT or Flattening Filter Free (FFF) beams. Experimental evidence confirms that the polymerization process in radiochromic films is widely insensitive to the instantaneous dose rate. Variations from conventionally delivered dose rates (e.g., 1 Gy/min) up to the extreme FLASH irradiation rates (e.g., thousands of Gy/s) have proven that the intrinsic RCF solid-state chemistry response remains unaffected by the rate of dose deposition.

5.3 Post-Irradiation OD Growth

A hallmark characteristic of radiochromic film involves the temporal dependency of its observable response. Unlike an instantaneous process, the chain-growth polymerization initiated by radiation does not cease the moment the beam is turned off. Instead, the polymer chains continue to propagate within the crystalline lattice, causing the film to progressively darken over time. This post-irradiation optical density (OD) growth operates roughly logarithmically with respect to time after the irradiation ceases. The rate of growth is initially rapid within the first 1 to 2 hours, before gradually plateauing. Although the change in OD after 24 hours is exceptionally slow ($< 1\%$ over the next few days), it conceptually never fully stops. Therefore, an essential dosimetric protocol is to ensure strict timing consistency: both the calibration films and the patient-specific measurement films MUST be read out (scanned) at the exact same delay following their respective irradiations (typically standardized to 24 hours).

6 Readout Systems and Data Acquisition Procedures

The fundamental signal in radiochromic film dosimetry is extracted via optical transmission—specifically, measuring the amount of light absorbed by the colored polymer chains. The data acquisition process requires converting the film darkness into an Optical Density (OD), mathematically defined as:

$$OD = \log_{10}\left(\frac{I_0}{I}\right) \tag{1}$$

where I_0 represents the initial intensity of the light source without the film, and I represents the transmitted or reflected intensity after the light intercepts the film. Various devices can act as readout systems, including point-based He-Ne laser densitometers and spectrophotometers. However, for 2D map extraction, high-resolution Color Flatbed Scanners have universally become the golden standard.

6.1 Color Flatbed Scanners

Commercial color flatbed document scanners, particularly models explicitly tested for dosimetric integrity such as the Epson Expression 10000XL, 11000XL, and 12000XL, are equipped with a linear array CCD (Charge-Coupled Device) combined with a broadband cold cathode fluorescent or LED light source. When paired with multi-channel dosimetry paradigms (like EBT3 film), flatbed scanners digitize the image by splitting the broad light spectrum into Red, Green, and Blue (RGB) color channels.

- **Red Channel:** Matches the main absorbance peak of the di-acetylene polymer ($\sim$635 nm) and is highly sensitive for low-dose regions (0 to $\sim$10 Gy).
- **Green Channel:** Is generally used to extend the dynamic range into higher doses (up to $\sim$40 Gy) as the red channel approaches saturation and becomes less responsive.
- **Blue Channel:** Frequently utilized not for direct dosimetry, but to mathematically map and correct spatial non-uniformities caused by minor variations in the active layer thickness of the film across its surface.

6.2 Differences Between Reflection and Transmission Scanning

Scanning protocols utilized in color flatbed densitometry generally operate under two fundamentally structurally distinct methodologies: Transmission scanning and Reflection scanning. Proper understanding of establishing the correct imaging geometry is critical for acquiring faithful dosimetric maps [1].

6.2.1 Transmission Scanning Mode

In transmission scanning, the scanner's primary physical setup relies on dual functionalities typically afforded by a Transparency Unit (TPU).

- **Mechanism:** The light source employed for scanning is situated in the upper lid of the scanner. As the scan progresses, light is transmitted directly downwards, traversing completely through the radiochromic film. The transmitted photons are subsequently captured by the CCD sensor array housed at the bottom scanning bed.

- **Applications and Advantages:** Transmission scanning is the absolute preferred and standard mode for translucent media such as radiographic and most clinical radiochromic films (e.g., EBT2, EBT3). Because the optical path passes exactly once through the dosimeter, it rigorously obeys the Beer-Lambert law of absorption, allowing for the most direct, artifact-free calculation of the true transmission Optical Density. It inherently provides a vast dynamic range, superior signal-to-noise ratio, and greater overall dosimetric accuracy.

6.2.2 Reflection Scanning Mode

Conversely, reflection scanning utilizes an entirely different photon path.

- **Mechanism:** In reflection mode, both the illumination light source and the CCD detector are located on the same side beneath the scanner glass bed. Light is emitted upwards, passes through the film, hits an opaque reflecting backing material (usually the white opaque document mat inside the scanner lid), and then is reflected back down. The light then passes through the film a second time before returning to the CCD sensor.
- **Applications and Disadvantages:** For relative dosimetry with EBT films, reflection scanning brings significant inherent degradation. Since light transverses the active polymer layer twice, the system suffers from an accelerated early saturation of the signal, drastically reducing the effective dynamic range. Furthermore, reflection is wildly susceptible to optical disturbances, surface scattering on the protective polyester layers of the RCF, and minute air gaps between the film and scanner glass, collectively translating to high noise levels and compromised quantitative reliability. While reflection scanning is mandatory for older opaque film models that do not permit light transmission, it is strongly discouraged for modern EBT films used in precision radiotherapy procedures.

7 Conclusion on Readout Optimization

By standardizing the timing of the readout stage—waiting consistently for the fast post-irradiation OD growth phase to stabilize (e.g., 24 hours)—and utilizing transmission mode with a multi-channel capable flatbed scanner, clinical medical physicists can extract highly accurate and intricately patterned planar dose distributions, confirming the geometric and dosimetric safety of highly modulated modern radiotherapy deliveries.

8 Observation

8.1 Calibration of known film doses

- **Machine:** Elekta Versa HD
- **SAD:** 100 cm
- **Buildup slab:** 5 cm
- **Base slabs:** 10 cm
- **PDD at 5 cm:** 0.86
- **Field Size (FS):** 10×10 cm^2, 6 MV photon

MU Calculation:

$$MU = \frac{Dose}{1\ \text{cGy}/MU \times PDD} \tag{2}$$

Dose and Corresponding Monitor Units (MU):

S.No.	Dose	MU
1	Background	-
2	0.2 Gy	23 MU
3	0.5 Gy	58 MU
4	1 Gy	116 MU
5	2 Gy	232 MU
6	3 Gy	349 MU
7	4 Gy	465 MU
8	5 Gy	581 MU
9	8 Gy	930 MU

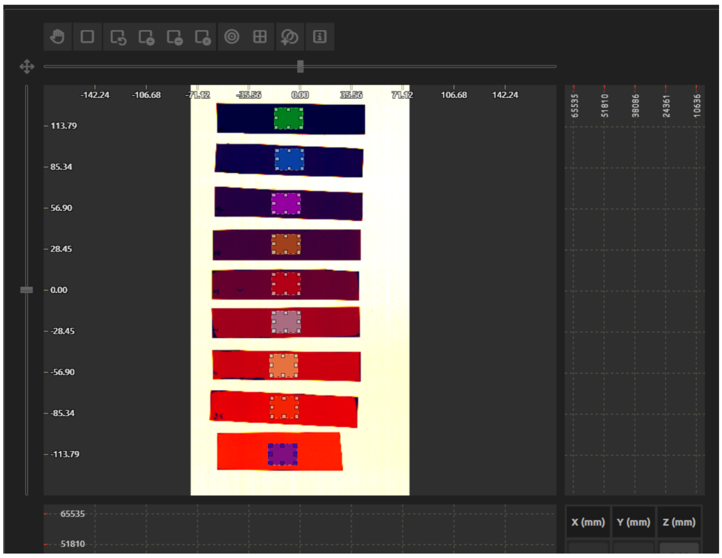
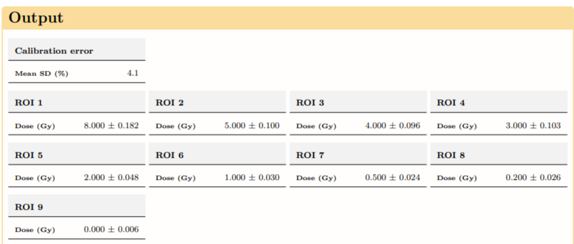


Figure 2: Film with known doses



Output

Calibration error	
Mean SD (%)	4.1

ROI 1		ROI 2		ROI 3		ROI 4	
Dose (Gy)	8.000 ± 0.182	Dose (Gy)	5.000 ± 0.100	Dose (Gy)	4.000 ± 0.096	Dose (Gy)	3.000 ± 0.103
ROI 5		**ROI 6**		**ROI 7**		**ROI 8**	
Dose (Gy)	2.000 ± 0.048	Dose (Gy)	1.000 ± 0.030	Dose (Gy)	0.500 ± 0.024	Dose (Gy)	0.200 ± 0.026
ROI 9							
Dose (Gy)	0.000 ± 0.006						

Figure 3: Calibration Output

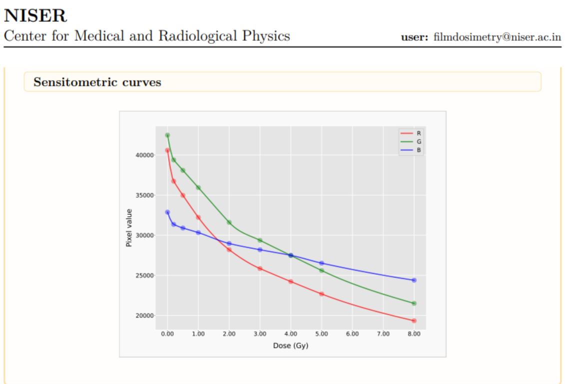


Figure 4: Sensitometric curves

8.2 Dose-Optical Density Relationship

Mathematically, it can be expressed as a negative logarithm of Transmittance:

$$\text{Optical Density} = -\log_{10}(T) = -\log_{10}\left(\frac{I}{I_0}\right) \quad \text{or} \quad \text{Optical Density} = \log_{10}\left(\frac{I_0}{I}\right) \tag{3}$$

Where T is transmittance, I_0 is the intensity of incident radiation, and I is the intensity of transmitted radiation.

Dose (Gy)	Pixel	OD
0	40493 (I_0)	0
0.2	36093	0.04995697197
0.5	34707	0.06696287769
1	31666	0.1067867461
2	27885	0.1620093047
3	25655	0.1982079345
4	24119	0.225020656
5	21969	0.2655696646
8	19023	0.3281009456

Table 1: Dose vs Pixel Value and Optical Density (OD)

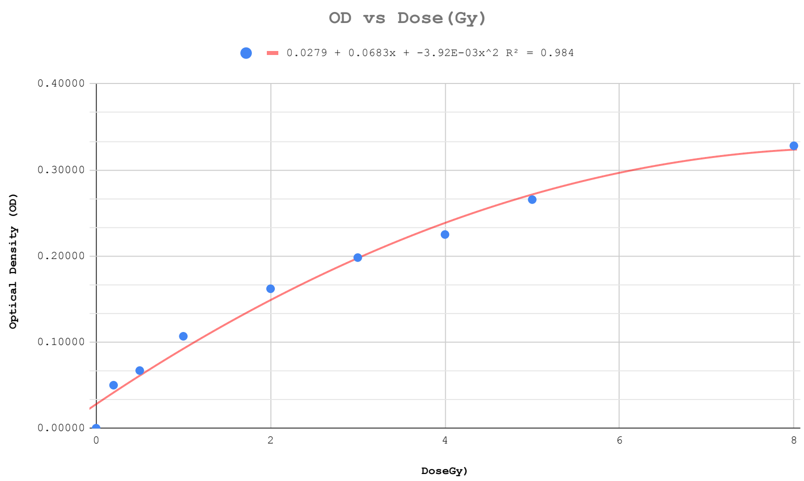


Figure 5: Dose vs Optical Density (OD)

8.3 Dosimetry of unknown doses

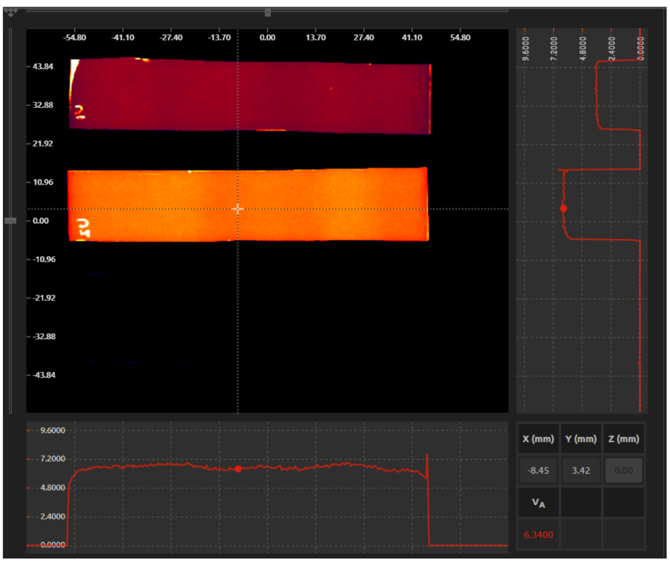


Figure 6: Film with unknown doses

8.3.1 Result

- Dose of Film 1 = 3.64 Gy
- Dose of Film 2 = 6.57 Gy

8.3.2 Error Analysis

Film	True Dose (Gy)	Calculated Dose (Gy)	Percentage Error (%)
Film 1	4.00	3.64	9.00
Film 2	6.90	6.57	4.78

Table 2: Error between calculated and true dose

8.4 Spoke Test

8.4.1 Couch Spoke Test

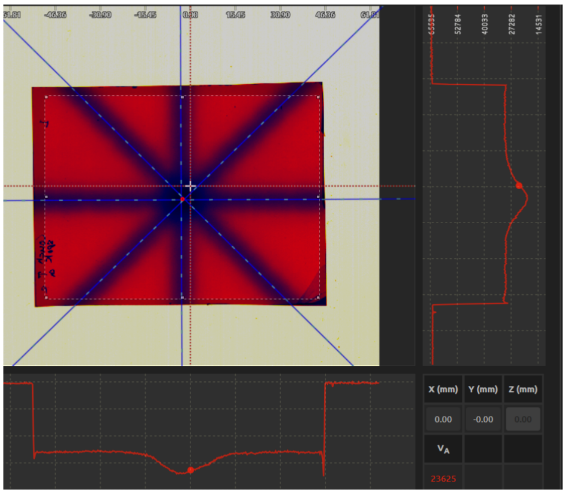


Figure 7: Couch Spoke Test

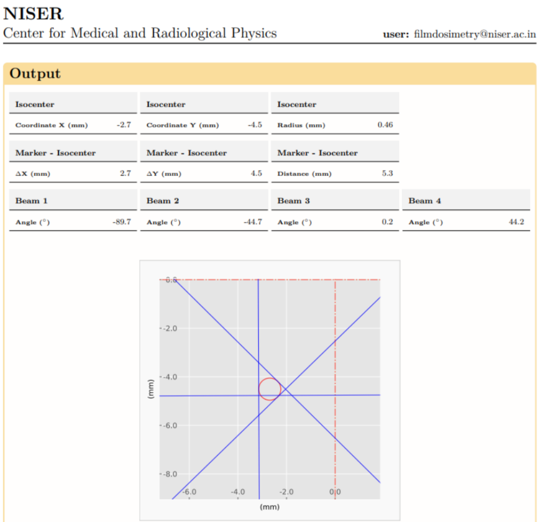


Figure 8: Couch Spoke Test Report

Isocenter Radius = 0.46mm, which is within tolerance(2mm).

8.4.2 Collimator Spoke Test

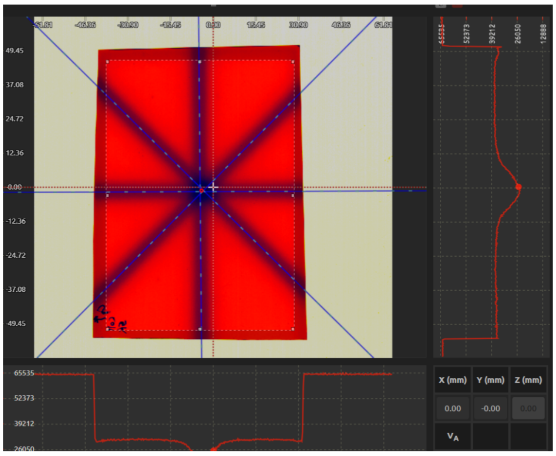
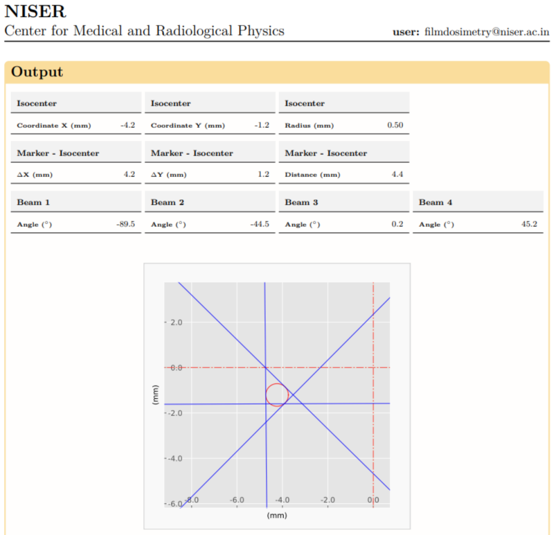


Figure 9: Collimator Spoke Test


NISER
Center for Medical and Radiological Physics — user: filmdosimetry@niser.ac.in

Output

Isocenter		Isocenter		Isocenter			
Coordinate X (mm)	-4.2	Coordinate Y (mm)	-1.2	Radius (mm)	0.50		
Marker - Isocenter		Marker - Isocenter		Marker - Isocenter			
ΔX (mm)	4.2	ΔY (mm)	1.2	Distance (mm)	4.4		
Beam 1		Beam 2		Beam 3		Beam 4	
Angle (°)	-89.5	Angle (°)	-44.5	Angle (°)	0.2	Angle (°)	45.2

Figure 10: Collimator Spoke Test Report

Isocenter Radius = 0.50mm, which is within tolerance(2mm)

8.4.3 Gantry Spoke Test

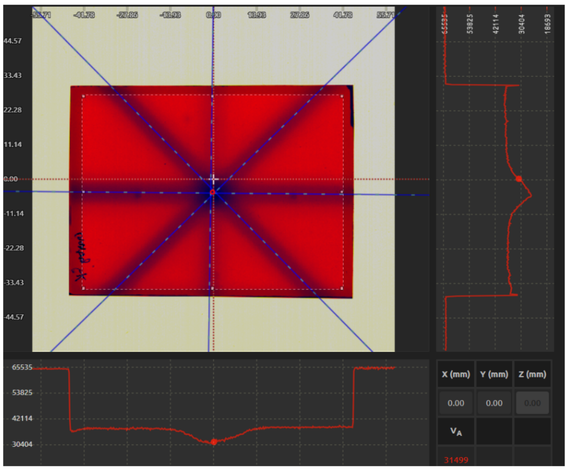
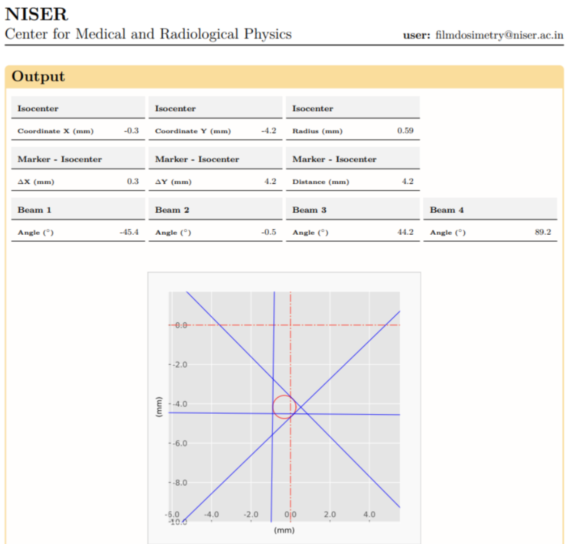


Figure 11: Gantry Spoke Test


NISER
Center for Medical and Radiological Physics — user: filmdosimetry@niser.ac.in

Output

Isocenter		Isocenter		Isocenter			
Coordinate X (mm)	-0.3	Coordinate Y (mm)	-4.2	Radius (mm)	0.59		
Marker - Isocenter		Marker - Isocenter		Marker - Isocenter			
ΔX (mm)	0.3	ΔY (mm)	4.2	Distance (mm)	4.2		
Beam 1		Beam 2		Beam 3		Beam 4	
Angle (°)	-45.4	Angle (°)	-0.5	Angle (°)	44.2	Angle (°)	89.2

Figure 12: Gantry Spoke Test Report

Isocenter Radius = 0.59mm, which is within tolerance(2mm)

8.5 PSQA Gamma Analysis

8.6 Field Size Congruence Test

8.6.1 Symmetric Field $5 \times 5\text{cm}^2$

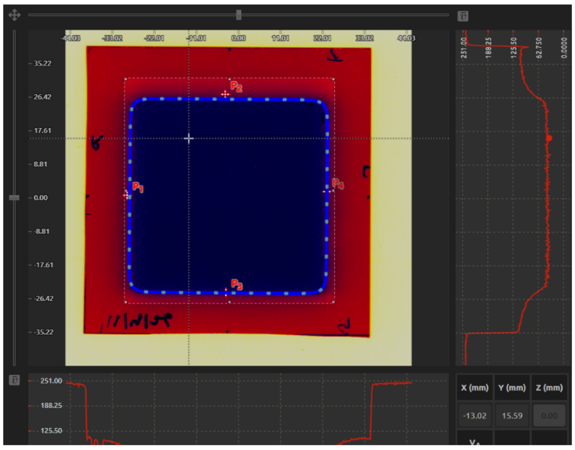
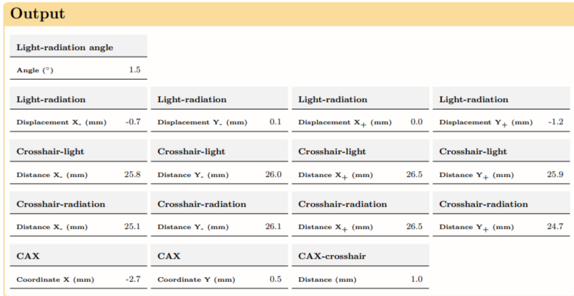


Figure 13: Symmetric Field



Output

Light-radiation angle	Angle (°)	1.5
Light-radiation	Displacement X_- (mm)	-0.7
Light-radiation	Displacement Y_- (mm)	0.1
Light-radiation	Displacement X_+ (mm)	0.0
Light-radiation	Displacement Y_+ (mm)	-1.2
Crosshair-light	Distance X_- (mm)	25.8
Crosshair-light	Distance Y_- (mm)	26.0
Crosshair-light	Distance X_+ (mm)	26.5
Crosshair-light	Distance Y_+ (mm)	25.9
Crosshair-radiation	Distance X_- (mm)	25.1
Crosshair-radiation	Distance Y_- (mm)	26.1
Crosshair-radiation	Distance X_+ (mm)	26.5
Crosshair-radiation	Distance Y_+ (mm)	24.7
CAX	Coordinate X (mm)	-2.7
CAX	Coordinate Y (mm)	0.5
CAX-crosshair	Distance (mm)	1.0

Figure 14: Symmetric Field Report

The Light-radiation displacement along the +x (in-line) and x (in-line) directions is 0 mm and -0.7 mm, respectively, while along the +y and y directions, it is -1.2 mm and 0.1 mm, respectively. As per AERB guidelines, the deviation should be less than ±2 mm.

8.6.2 Asymmetric Field $5 \times 8\text{cm}^2$

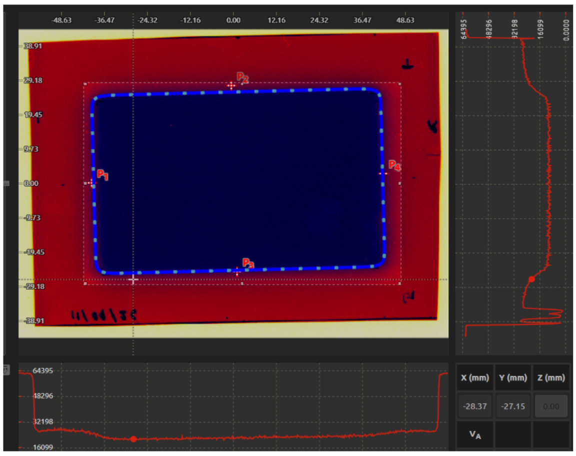
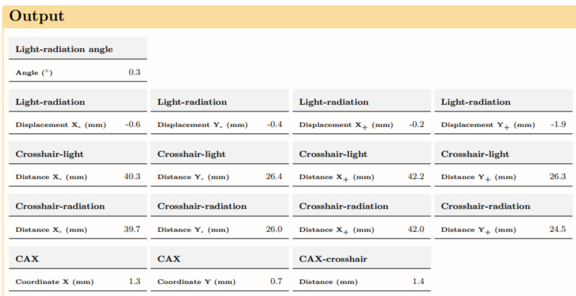


Figure 15: Asymmetric Field



Output

Light-radiation angle	Angle (°)	0.3
Light-radiation	Displacement X_- (mm)	-0.6
Light-radiation	Displacement Y_- (mm)	-0.4
Light-radiation	Displacement X_+ (mm)	-0.2
Light-radiation	Displacement Y_+ (mm)	-1.9
Crosshair-light	Distance X_- (mm)	40.3
Crosshair-light	Distance Y_- (mm)	26.4
Crosshair-light	Distance X_+ (mm)	42.2
Crosshair-light	Distance Y_+ (mm)	26.3
Crosshair-radiation	Distance X_- (mm)	39.7
Crosshair-radiation	Distance Y_- (mm)	26.0
Crosshair-radiation	Distance X_+ (mm)	42.0
Crosshair-radiation	Distance Y_+ (mm)	24.5
CAX	Coordinate X (mm)	1.3
CAX	Coordinate Y (mm)	0.7
CAX-crosshair	Distance (mm)	1.4

Figure 16: Asymmetric Field Report

The Light-radiation displacement along the +x (in-line) and x (in-line) directions is -0.2 mm and -0.6 mm, respectively, while along the +y and y directions, it is -1.9 mm and -0.4 mm, respectively. As per AERB guide- lines, the deviation should be less than ±2 mm

8.7 Radiation Field Analysis(Symmetric Field $5 \times 5\text{cm}^2$)

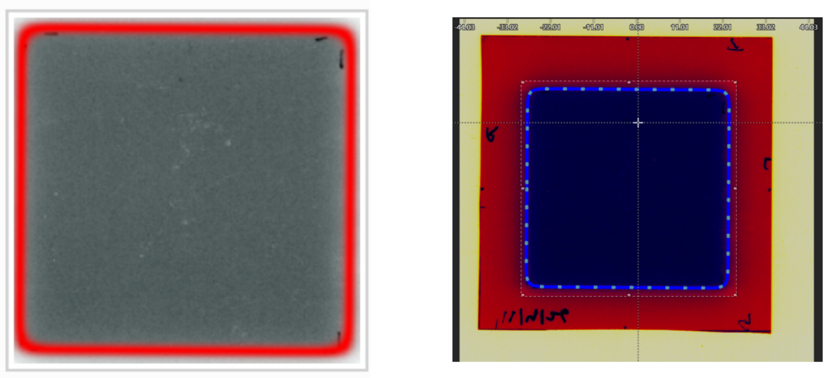


Figure 17: Radiation Field Analysis

Output

Radiation field center	Coordinate X (mm)	-2.3
Radiation field center	Coordinate Y (mm)	0.6
Radiation field size	X (mm)	51.6
Radiation field size	Y (mm)	50.8
Field rotation	Applied rotation (°)	-0.4
Central dose statistics	Measuring points	28
Central dose statistics	Mean	40.57
Central dose statistics	Median	41.00
Central dose statistics	Standard deviation	2.27
Penumbra	X_- (mm)	3.2
Penumbra	Y_- (mm)	2.4
Penumbra	X_+ (mm)	3.0
Penumbra	Y_+ (mm)	2.3
Flatness	X (%)	1.5
Flatness	Y (%)	2.1
Flatness	Diagonal_1 (%)	2.5
Flatness	Diagonal_2 (%)	3.3
Symmetry	X (%)	-1.6
Symmetry	Y (%)	4.0
Symmetry	Diagonal_1 (%)	4.7
Symmetry	Diagonal_2 (%)	-5.4

Figure 18: Radiation Field Analysis Report

9 Conclusion

A comprehensive understanding of the governing physical principles and stringent Quality Assurance protocols ensures optimal clinical outcomes and fundamental radiation safety.

References

[1] A. Niroomand-Rad, S.-T. Chiu-Tsao, M. P. Grams, D. F. Lewis, C. G. Soares, L. J. Van Battum, I. J. Das, S. Trichter, M. W. Kissick, G. Massillon-JL, P. E. Alvarez, and M. F. Chan, "Report of aapm task group 235 radiochromic film dosimetry: An update to tg-55," *Medical Physics*, vol. 47, no. 12, pp. 5986–6025, 2020.